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Refining Abstractive Text Summarization: Towards an Optimizing Approach based on PTLM Selection and Fine-Tuning

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***Abstract*—This study provides an out-of-the-box inferential and empirical analysis of four state-of-the-art large natural language models namely, BART, PEGASUS, ProphetNet, and T5, assessing their summarization capabilities on three diverse datasets, CNN/DailyMail, Gigaword, and XSUM. The analysis uniquely positions itself by considering the models' performance without any task-specific fine-tuning, the results are quantified using the ROUGE metric, offering insights into each model's ability to generate coherent and concise summaries directly out-of-the-box. The results indicated significant variations in performance, with BART and PEGASUS generally leading in terms of ROUGE scores and actual summary inspection, T5 excelling in generating concise summaries with limited information and mainly being extractive (copying), and ProphetNet showing a need for potential task-specific adjustments. The findings offer guidance for selecting and deploying pre-trained language models for real-world summarization applications within a low-resource environment.**

***Index Terms*—PLTM, abstractive text summarization, natural language processing, large language models**

I. Introduction

The era of big data has given rise to generation and acquisition of varieties of data from different sources. One of the common and voluminous kinds of data is text data which is generated from sources such as customer inquiries, social media comments and posts, weblogs, online books, and scientific journals. With the explosion of text data, it becomes essential to develop effective Automatic Text Summarization (ATS) techniques. ATS techniques take text as input and produce a shorter form or version of the input text while maintaining its original meaning and context. Existing literature classifies ATS techniques into extractive and abstractive methods [1].

Abstractive text summarization produces a novel summary without reference to the words and sentences from the original document while maintaining the context of the input document [2], and has continuously solved summarization issues that include coherence, readability, and redundancy, which are salient to the extractive method. Abstractive text summarization is more desirable because of its capabilities and quality of summary generated. Since the advent of abstractive text summarization, this area has been fraught with challenges such as identifying the most significant segments in the input text for inclusion in the summary, summarizing lengthy documents like books, multi-document summarization, and creating abstract summaries comparable to those produced by humans.

Abstractive text summarization has evolved through the use of non-neural statistical approaches, machine learning, Deep Sequence-to-Sequence, and Pre-Trained Language Models (PTLMs) [3]. PTLMs such as GPT-3, GPT-4, BERT, RoBERTa and BART, have achieved state-of-the-art (SOTA) results in NLP problems and have shown great opportunities for further improvements [4]. Using variants and forms of PTLMs for abstractive text summarization, the generative NLP field has seen several improvements and has recorded successes. Having said that, one of the challenging considerations while using PTLMs is selecting an appropriate model for a downstream task [5].

In view of the challenges identified, the contributions of this paper are as follows: (1) Performs comparative performance analysis of state-of-the-art PLTMs (BART, PEGASUS, ProphetNet, and T5) on three standard datasets (CNN/DailyMail, Gigaword, and XSUM) (2) Presents a deeper understanding the roles of pre-training objective in out-the-box implementation of PTLM in no or low resource environment. (3) Establishes the need for an understanding the dataset, awareness of PTLM extractive bias when not fine-tuned and the importance of fine-tuning for a downstream task.

The rest of the paper is organized in the following manner: Section II provides a review of the use of PLTMs for abstractive text summarization. Section III elaborates on the methodology used for the experimental comparison performed in this work. Section IV summarizes and provide an analysis of results. Finally, Section V synopsizes the findings.

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II. Pre-Trained Language Models for Abstractive Text Summarization

Deep Sequence-to-Sequence techniques such as LSTM and GRU, showed advancement over the statistic and rule-based techniques [3]. Unlike traditional machine learning and other non-neuron-based techniques, they are automatic, and researchers don't need to extract features manually. However, due to their recurrent nature and sequential procedure, they failed to model long-region dependencies and are unable to parallelize sequence processing [5] leading to a loss of meaningful information in summarization and slow processing.

An effort to solve the parallelism issue of RNNs ushered in the use of Convolutional Neural Network (CNN), which is able to parallelize sequence processing but do not work well with very long documents and are unable to solve long-term dependency issues well. To solve these issues, [6] provided transformer-based models with their self-attention mechanism which allows the models to focus on different parts of the input sequence and learn their associated relationships regardless of their positions in the input text thereby solving the dependency issues and helping to improve the generated summary. They outperformed RNNs and CNN-based approaches.

The quadratic memory issue, which arises as the memory requirements increase quadratically with the size of the input document and other text summarization problems in the previous methods were solved using Pre-Trained Language Models (PTLM) which achieved SOTA outcomes in abstractive summarization [5]. PTLMs are trained on a vast corpus of text and learn to predict the next word in a sentence, which allows them to generate fluent natural language text when perfectly adapted to a downstream task or specific dataset.

The inception of Pre-Trained Language Models (PTLMs) has dramatically transformed numerous sectors within natural language processing (NLP), specifically in the field of Generative AI and Abstractive Text Summarization (ATS). Most PTLMs use transformer models as their core architecture, while abstractive ATS applications are developed using PTLMs by harnessing the principles of transfer learning. PTLMs are language models trained on huge amounts of unlabelled text data and have gained a good understanding of language patterns, syntax, and semantics, exploiting the parallel processing capabilities of transformers to expedite the training process.

This leads to the creation of universal language representations that can be applied, or transferred, to downstream tasks such as abstractive text summarization, thereby enhancing their performance and reducing their training time. Once pre-trained, these models can be fine-tuned on a specific task with a relatively small amount of task-specific data [7]. This methodology has led to state-of-the-art results on a wide variety of NLP tasks. In light of abstractive ATS, popular PTLMs including PEGASUS, ProphetNet, T5, and BART have proven effective [8]. Therefore, the four models chosen for this study include T5, BART, ProphetNet and PEGASUS. These models have been detailed below.

A. Text-to-Text Transfer Transformer (T5)

The T5 (Text-to-Text Transfer Transformer) model represents a novel approach in natural language processing (NLP), treating every NLP task as a text-to-text problem [9]. This unifying concept frames diverse tasks such as translation, summarization, question answering, and classification as converting one form of text to another, simplifying the architecture and training process of the model. T5's core training involves span masking, where it masks contiguous spans of text, unlike traditional models that mask individual words. This method compels the model to predict not just isolated words but entire phrases or sentences, thereby enhancing its understanding of context and language nuances.

One of the key benefits of T5's approach is its flexibility and simplicity, allowing it to adapt to various NLP tasks with a single unified model architecture. The span prediction training tasks, coupled with diverse text-to-text conversions, enable T5 to develop a comprehensive understanding of language context. This training strategy helps the model generalize well across different types of NLP tasks. However, the model might face challenges in addressing specific nuances unique to individual tasks, given its generalized approach. Overall, T5's methodology marks a significant step in creating versatile and effective NLP models.

B. Bidirectional and Auto-Regressive Transformers (BART)

BART is a novel denoising autoencoder for pre-training sequence-to-sequence models, which stands out for its remarkable text generation capabilities. It combines a bidirectional encoder and a left-to-right autoregressive decoder, employing the Transformer architecture as its backbone, while introducing variations such as GeLU activations and a specific parameter initialization strategy [10].

The model is pre-trained using a distinctive approach that involves corrupting text with several noising methods, such as token masking, token deletion, text infilling, sentence permutation, and document rotation, and then learning to reconstruct the original text [11]. This pre-training regime enables BART to handle a variety of text alterations and enhances its ability to infer and reconstruct missing information, making it robust and versatile summarization as seen in its out-of-box performance on the challenging XSum dataset.

C. ProphetNet

While both the original Transformer and ProphetNet are rooted in the Transformer encoder-decoder architecture, ProphetNet introduces several innovations including the future n-gram prediction [8] and Mask Based Autoencoder Denoising Pre-Training Objectives, and the multiple self-attention streams, which all set it apart from the original Transformer design.

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This model integrates extra self-attention prediction streams where each stream is tasked with predicting a specific future token. While the primary ”stream” works exactly like the standard Transformer’s approach, extra streams predict subsequent tokens based on prior stream’s states. ProphetNet also introduces a secondary predicting stream with unique initialization tokens and positional calculations [5]. This allows for a seamless transition to a standard Transformer decoder by deactivating the predicting streams during inference.

Additionally, this model uses Mask Based Autoencoder Denoising Pre-Training Objective, which is an added twist for Seq2Seq pre-training. It involves an autoencoder, a neural network that learns to introduce noise into the data and then reconstruct (denoise) it. The corruption is done by ”masking” (replacing words with a [MASK] token) parts of the input. For ”denoising”, the model is trained to recover original data from a corrupted version, helping it focus on essential features and ignore noise. The model tries to predict these masked parts using the visible data. Unlike its peers, ProphetNet restores several future tokens within each masked section. This task enables the model to grasp the data’s context and significance, enhancing its performance in generation tasks such as summarization or question generation.

D. PEGASUS

Pre-training with Extracted Gap-sentences for Abstractive SUmmarization Sequence-to-sequence or PEGASUS is pre-trained specifically for abstractive text summarization by selecting sentences from the document to form a pseudo-summary. The PEGASUS model introduces a novel pre-training approach for large Transformer-based encoder-decoder models to improve abstractive summarization. Instead of traditional methods like Masked Language Modeling (MLM) where individual words or spans are masked, PEGASUS masks entire sentences in the input document [8], [11]. These masked sentences are chosen based on their importance to the document, mimicking an extractive summary. The model is then trained to generate these sentences as a single output sequence.

The Gap Sentences Generation (GSG) task involves different strategies to select sentences to mask, such as selecting the lead sentences, randomly selecting sentences, or selecting principal sentences based on their ROUGE-1 score against the rest of the document. Among these, the ”Ind-Orig” strategy, which selects independent principal sentences, showed the best performance [11]. The Gap Sentences Ratio (GSR), indicating the proportion of sentences masked, is also explored, with 30% found to be an effective ratio for the PEGASUSLARGE model.

III. Methodology

This study includes inferential comparison of the four state-of-the-art language models (T5, BART, ProphetNet, and PEGASUS) on the XSum, CNN/DailyMail, and the Gigaword datasets. Therefore, this section outlines the experimental process used for evaluating the performance of the four text summarization models on three datasets.

A. Experimental Setup

The development and programming environment involved using the Hugging Face platform and Google Colab as the primary resources. The Hugging Face platform was used to access pre-trained models and datasets. On the other hand, Google Colab is a cloud-based service, which provides a Jupyter Notebook Python environment with access to computing resources. The experiment utilized the free Google Colab’s computing resources option, which makes the process independent of Personal Computer (PC) capabilities. The free tier offers 2 x vCPU (Virtual CPU) instances and 13 GB of RAM.

As this is an inferential experiment, there was no need to fine-tune or train any model from scratch. Therefore, the study does not explore training or fine-tuning concepts such as variable number of epoch(s) for training. The study aims to establish a case for performing fine-tuning and these aspects will be explored in future work. The steps involved in performing this experimentation are listed below -

1) The installation of transformers, sentencepiece, and datasets with updates made to ensure compatibility within the Google Colab environment.
2) The dataset structures, including splits and features, were explored.
3) A baseline was established by extracting the first three sentences of a sample article.
4) Summaries were generated using the T5, BART, ProphetNet, and PEGASUS models, each configured and executed within the environment.
5) Generated summaries were compared against reference summaries from the datasets using the ROUGE scoring metric.

A specific piece of text from the dataset, truncated to 2000 characters, was prepared for summarization input to manage computational resources effectively and standardize input size across tests. Besides this, for fair comparison large-scale model checkpoints (‘google/flan-t5-large‘, ‘facebook/bart-large‘, ‘microsoft/prophetnet-large-uncased‘, and ‘google/pegasus-large‘) were selected to ensure a level playing field based on model scale, capacity, and architectural representation.

B. Datasets

To achieve a robust comparison, model performances were analysed across three distinct datasets—CNN/DailyMail, Gigaword, and XSUM, each representing different summarization challenges. The CNN/DailyMail dataset tests the model’s ability to extract and synthesize key points from lengthy documents, Gigaword challenges models to be succinct and catchy, and XSUM requires a high degree of information compression

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and abstraction. These distinctive goals or behaviours of the datasets with different models, and the models' performance provides insights into their versatility and practical utility.

1) CNN/DailyMail Dataset: The CNN/DM dataset, modified for abstractive automatic text summarization (ATS), contains news articles from CNN and The Daily Mail, each paired with multi-sentence summaries [5]. It has two versions: anonymized and non-anonymized. The dataset comprises of 287,227 training pairs, 13,368 validation pairs, and 11,490 test pairs. Each article averages around 700 tokens, with summaries averaging 3-4 sentences.

The CNN/DailyMail dataset is a large collection of news articles originally curated for machine learning research on text summarization. It includes articles along with multiple human-written summaries (highlights). The articles are long-form, often containing detailed narratives and a complex structure that require the model to understand and condense effectively. The objective is to produce a summary that captures the essential points of the article. This typically involves multi-sentence outputs that reflect the key facts and arguments presented in the source text. The CNN/DailyMail dataset requires models to condense lengthy news articles into shorter versions.

2) Gigaword Dataset: Gigaword is a dataset compiled from various news sources, primarily used for the task of headline generation. Unlike the CNN/DailyMail dataset, the summarization task here is to generate a single, concise headline that captures the main point of the news article. It consists of 4 million article-title pairs, primarily aimed at generating headlines from the first sentence of an article [5], [11]. It gathers content from seven news sources, including Agence France-Presse, Associated Press, Central News Agency of Taiwan, Los Angeles Times/Washington Post, Washington Post/Bloomberg, New York Times, and Xinhua News Agency. Despite its extensive collection, the Gigaword corpus is less recognised in modern abstractive automatic text summarization research, which typically focuses on summarizing multiple sentences rather than single-sentence headline generation.

One of the challenges of this dataset is the brevity as headlines must be very short, usually under 10 words, forcing models to distil the essence of an article into a very compact form. Other challenges include creativity and keyword emphasis. Effective headlines often include plays on words, puns, or other creative language use that can be challenging for models to replicate. Meanwhile, headlines must include critical keywords that convey the core subject matter of the article, which requires the model to prioritize information differently than in longer summaries.

3) XSum: The XSum dataset, designed to enhance the novelty in abstractive ATS models, consists of BBC articles with single-sentence summaries. It includes 90% training pairs and 5% each for validation and test pairs. The articles average around 20 sentences with 400 tokens, while the summaries are about 23 tokens long, answering what is the article about. The dataset spans a wide range of topics, requiring models to be adaptable across different subjects and writing styles [5].

The goal is to generate a single sentence that captures the most salient point of the article. The single-sentence summary requires not just condensing but also synthesizing information, often necessitating the model to fill in gaps with its understanding, [10] and also requires the model to construct novel sentences that are not found in the text, testing its ability to generate coherent and contextually appropriate content.

C. Evaluation Metric

The evaluation metric used for comparison is ROUGE (Recall-Oriented Understudy for Gisting Evaluation), which compares the machine-generated summaries with reference summaries in the test portion of each dataset. ROUGE, the commonly used metric [8], is a collection of metrics used to evaluate the quality of text summaries by comparing them with reference summaries, usually created by humans. It focuses on the overlap of content between the machine-generated summary and the reference, with different variants of ROUGE measuring different aspects of this overlap.

ROUGE-N captures the overlap of n-grams (e.g., unigrams, bigrams) between the system and reference summaries, reflecting the presence of key phrases. N can take the values 1, 2, and so on. Dong et al. Extractive summarization systems often score higher on ROUGE than abstractive systems. This is partly because news articles usually have crucial information at the beginning, making simple strategies like selecting the initial content effective. ROUGE's design, favouring exact word matches, tends to reward these safer strategies, making it challenging for abstractive systems to score high. Literature points out that having only one reference summary may reduce ROUGE's reliability, as it may not capture the diversity of valid summaries [12].

Other types of ROUGE scores include ROUGE-L, ROUGE-S and ROUGE-W. ROUGE-L measures the longest common subsequence of words, indicating the fluency and order of the generated text. On the other hand, ROUGE-S looks at skip-bigrams to assess the relationship between non-consecutive word pairs in the summary. Lastly, ROUGE-W weighs longer matching sequences more heavily, rewarding more coherent longer phrases. This study uses ROUGE-1, ROUGE-2 and ROUGE-L for comparative evaluation.

IV. Results

This section presents a comparison of ROUGE scores for different models across three datasets: XSUM, Gigaword, and CNN/DailyMail, which is tabulated in Table I. The columns represent ROUGE1 (R1), ROUGE-2 (R2), and ROUGE-L (RL) scores. The index lists the models PEGASUS, ProphetNet, BART, and T5. Bold values highlight the highest scores for each dataset: T5 score d highest on Gigaword with 0.53 for ROUGE-L, BART scored highest on CNN/DailyMail with

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TABLE I
SUMMARY OF RESULTS

Models	XSUM			Gigaword			CNN/DailyMail		
	ROUGE-1	*ROUGE-2*	*ROUGE-L*	*ROUGE-1*	*ROUGE-2*	*ROUGE-L*	*ROUGE-1*	*ROUGE-2*	*ROUGE-L*
PEGASUS	**0.35**	0	0.24	0.17	0	0.17	0.21	0.04	0.14
ProphetNet	0	0	0	0	0	0	0.29	0.15	0.19
BART	0.32	0.06	0.17	0.17	0	0.17	**0.37**	0.14	0.26
T5	0.34	0.07	0.28	**0.53**	0.15	**0.53**	0.22	0.08	0.29

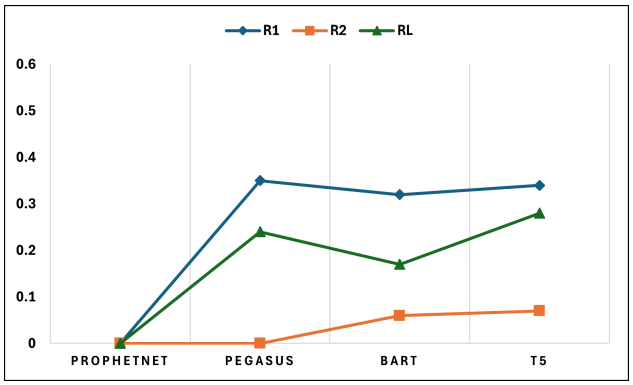

Fig. 1. Comparison of ROUGE scores for CNN/DailyMail Dataset

0.37 for ROUGE-L, and PEGASUS scored highest on XSUM with 0.35 for ROUGE-1.

Generally, the table suggests that T5 performs best on the Gigaword dataset, BART excels on the CNN/DailyMail dataset, and PEGASUS leads on the XSUM dataset in terms of the ROUGE-1 score. It also shows the unprecedented poor performance for the ProphetNet. Overall, the choice of the best model seems to depend on the dataset, with Pegasus standing out on XSUM and BART doing quite well on CNN/Daily Mail. Analysis of the results suggests that variability could be due to differences in the size, style, or subject matter of the datasets.

A. Performance on CNN/DailyMail Dataset

The CNN/DailyMail dataset, characterized by well-structured articles and detailed multi-sentence summaries, posed a challenging task for the evaluated models. Fig. 1 provides a comparison of the ROUGE scores for this dataset. BART distinguished itself with its uniform performance across various ROUGE metrics, while other models struggled. This consistency suggests that BART's pre-training for denoising text is particularly effective in handling the complexity of this dataset. In contrast, T5 and ProphetNet demonstrated limited proficiency in generating multi-sentence summaries, failing to incorporate a broader range of content from the articles with ProphetNet showing the worst performance in terms of the actual summary text, which is completely off context.

The ROUGE-1 score for ProphetNet indicates that a few sequences overlap, which again demonstrate the ProphetNet's inability to perform without fine-tuning. PEGASUS, while recognized for its capability in multi-sentence generation, surprisingly under-performed on this dataset. The Baseline method, comprising the first three sentences of the articles, achieved high ROUGE scores. This outcome indicates a potential bias in the ROUGE metric towards extractive summarization, as the Baseline's success stems from significant word overlap with the original articles and the subsequently the highlight (summary/gold standard). This also explains BART's high scores, suggesting its little extractive nature when not specifically tuned for abstraction, although with relatively higher coherence when compared to other models.

B. Performance on Gigaword Dataset

The Gigaword dataset, focusing on generating headlines from single sentences, presented a unique challenge. As is evident from Fig. 2, T5 excelled, most likely due to its text-to-text framework, which is adept at condensing information into concise summaries. Conversely, BART and PEGASUS, perhaps due to their strengths in multi-sentence summarization, struggled with this short sequence data task, potentially owing to their pre-training goals not aligning with the brevity required for headline creation. While PEGASUS and BART produced meaningful summaries, their failure to score high ROUGE value could be attributed to their lack of word overlap with the target headlines, which hindered their performance under the ROUGE metric. It is expected for PEGASUS and

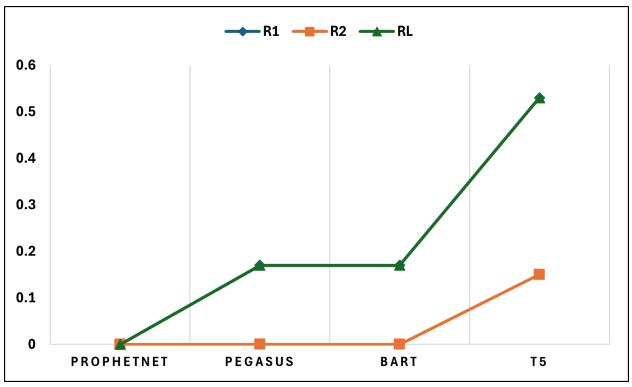

Fig. 2. Comparison of ROUGE scores for Gigaword Dataset
*Please note that the trendlines for R1 and RL are the same in the graph and thus, only one green trendline is visible.

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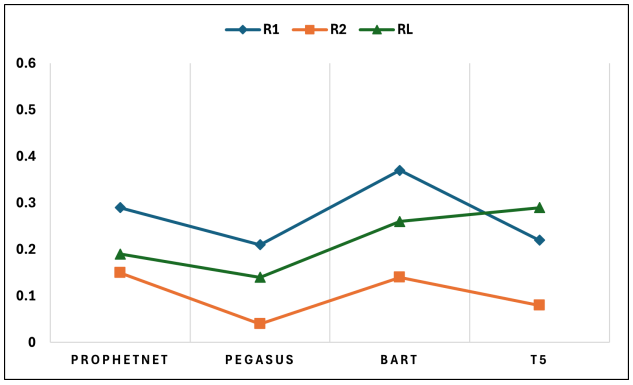

Fig. 3. Comparison of ROUGE scores for XSUM Dataset

BART to be outstanding on Gigaword dataset reading through their respective summaries, but were at the disadvantage of the ROUGE metric, which cares more about word/gram overlaps than meaningful summary, which explains an unprecedented high rouge score for T5.

C. Performance on XSUM Dataset

The XSUM dataset required a single-sentence summary describing an entire article, which is a test of abstraction and succinctness. Table I and Fig. 3 indicate that PEGASUS and BART showed similar ROUGE scores on this dataset because the dataset encouraged them to demonstrate their abstractive capacity. The ROUGE score for PEGASUS, BART, and T5 on this occasion are comparably close. As expected, the ROUGE values are close because the actual summaries all seem to have copied one or more words from the original document, but in a paraphrased manner in the case of PEGASUS and BART. A close look at T5 suggests highest number of consecutive overlapped gram, which makes it a better extractive candidate when not fine-tuned. Again, ProphetNet demonstrated weakness to generate summary when the data is small and when not fine-tuned.

V. Conclusion

The results across different summarization models indicate that model performance is highly contingent on dataset specificity and the inherent pre-training of each model. It is important to say that a PTLM behaves in an extractive manner when not fine-tuned to a downstream dataset (task). However, because they have been pre-trained on large corpus, they may create a paraphrased extractive summary. This approach produced little coherence, and meaningful summary as opposed to standard extraction. ProphetNet's poor performance across the datasets suggest the model's inability to generate any summary, single or multi-sentence, without fine-tuning. It is worth mentioning that ROUGE metric investigation is not a focus of this report, its impacts on the outcome of the analysis cannot be overlooked. ROUGE is deemed biased towards word overlap between the summary and original document (extractive). Thus, the ROUGE metric may not be the most accurate indicator of a model's capacity for abstractive summarization. This work suggests that BART and PEGASUS can be taken forward as fine-tuning candidates given their performances with future work also focusing on the identification of an effective metric for evaluation.

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References

[1] Widyassari, A.P., Rustad, S., Shidik, G.F., Noersasongko, E., Syukur, A. and Affandy, A., 2022. Review of automatic text summarization techniques & methods. Journal of King Saud University-Computer and Information Sciences, 34(4), pp.1029-1046.

[2] Gambhir, M. and Gupta, V., 2017. Recent automatic text summarization techniques: a survey. Artificial Intelligence Review, 47, pp.1-66.

[3] Syed, A.A., Gaol, F.L. and Matsuo, T., 2021. A survey of the state-of-the-art models in neural abstractive text summarization. IEEE Access, 9, pp.13248-13265.

[4] Ghadimi, A. and Beigy, H., 2022. Hybrid multi-document summarization using pretrained language models. Expert Systems with Applications, 192, p.116292.

[5] Alomari, A., Idris, N., Sabri, A.Q.M. and Alsmadi, I., 2022. Deep reinforcement and transfer learning for abstractive text summarization: A review. Computer Speech & Language, 71, p.101276.

[6] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is all you need. Advances in neural information processing systems, 30.

[7] Yu, Y., Zhang, R., Xu, R., Zhang, J., Shen, J. and Zhang, C., 2022. Cold-start data selection for few-shot language model fine-tuning: A prompt-based uncertainty propagation approach. arXiv preprint arXiv:2209.06995.

[8] Ahuir, V., Hurtado, L.F., González, J.Á. and Segarra, E., 2021. Nasca and nases: Two monolingual pre-trained models for abstractive summarization in catalan and spanish. Applied Sciences, 11(21), p.9872.

[9] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W. and Liu, P.J., 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. The Journal of Machine Learning Research, 21(1), pp.5485-5551

[10] Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V. and Zettlemoyer, L., 2019. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. arXiv preprint arXiv:1910.13461.

[11] Zhang, J., Zhao, Y., Saleh, M. and Liu, P., 2020, November. Pegasus: Pre-training with extracted gap-sentences for abstractive summarization. In International Conference on Machine Learning (pp. 11328-11339).

[12] See, A., Liu, P.J. and Manning, C.D., 2017. Get to the point: Summarization with pointer-generator networks. arXiv preprint arXiv:1704.04368.

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